

Electrical Circuits

Lec 5: Capacitors and Inductors

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Book

- **Fundamentals of Electric Circuits**
 - Charles K. Alexander
 - Matthew N. O. Sadiku
- 5th edition

Capacitors

- A capacitor is a passive element designed to store energy in its electric field.
- A **capacitor** consists of two conducting plates separated by an insulator (or dielectric).

$$q = Cv$$

$$C = \frac{\epsilon A}{d}$$

- **Capacitance** is the ratio of the charge on one plate of a capacitor to the voltage difference between the two plates, measured in farads (F).

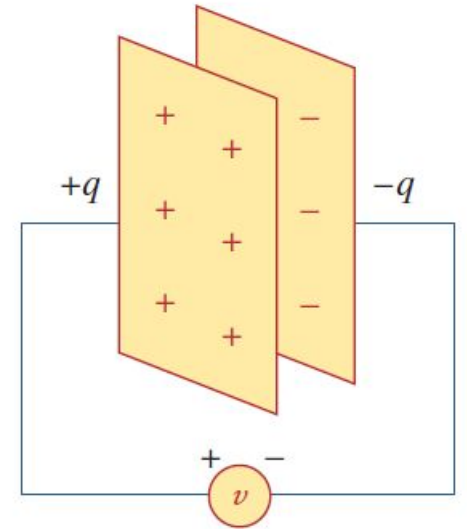


Figure 6.2

A capacitor with applied voltage v .

To obtain the current-voltage relationship of the capacitor, we take the derivative of both sides of Eq. (6.1). Since

$$i = \frac{dq}{dt} \quad (6.3)$$

differentiating both sides of Eq. (6.1) gives

$$\boxed{i = C \frac{dv}{dt}} \quad (6.4)$$

The voltage-current relation of the capacitor can be obtained by integrating both sides of Eq. (6.4). We get

$$v(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau \quad (6.5)$$

or

$$v(t) = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0) \quad (6.6)$$

The instantaneous power delivered to the capacitor is

$$p = vi = Cv \frac{dv}{dt} \quad (6.7)$$

The energy stored in the capacitor is therefore

$$w = \int_{-\infty}^t p(\tau) d\tau = C \int_{-\infty}^t v \frac{dv}{d\tau} d\tau = C \int_{v(-\infty)}^{v(t)} v dv = \frac{1}{2} Cv^2 \Big|_{v(-\infty)}^{v(t)} \quad (6.8)$$

We note that $v(-\infty) = 0$, because the capacitor was uncharged at $t = -\infty$. Thus,

$$w = \frac{1}{2} Cv^2$$

(6.9)

Using Eq. (6.1), we may rewrite Eq. (6.9) as

$$w = \frac{q^2}{2C} \quad (6.10)$$

Example 6.1

- (a) Calculate the charge stored on a 3-pF capacitor with 20 V across it.
- (b) Find the energy stored in the capacitor.

Solution:

- (a) Since $q = Cv$,

$$q = 3 \times 10^{-12} \times 20 = 60 \text{ pC}$$

- (b) The energy stored is

$$w = \frac{1}{2}Cv^2 = \frac{1}{2} \times 3 \times 10^{-12} \times 400 = 600 \text{ pJ}$$

Practice Problem 6.1

What is the voltage across a $4.5\text{-}\mu\text{F}$ capacitor if the charge on one plate is 0.12 mC ? How much energy is stored?

Answer: 26.67 V , 1.6 mJ .

Example 6.2

The voltage across a $5\text{-}\mu\text{F}$ capacitor is

$$v(t) = 10 \cos 6000t \text{ V}$$

Calculate the current through it.

Solution:

By definition, the current is

$$\begin{aligned} i(t) &= C \frac{dv}{dt} = 5 \times 10^{-6} \frac{d}{dt}(10 \cos 6000t) \\ &= -5 \times 10^{-6} \times 6000 \times 10 \sin 6000t = -0.3 \sin 6000t \text{ A} \end{aligned}$$

Example 6.3

Determine the voltage across a $2\text{-}\mu\text{F}$ capacitor if the current through it is

$$i(t) = 6e^{-3000t} \text{ mA}$$

Assume that the initial capacitor voltage is zero.

Solution:

Since $v = \frac{1}{C} \int_0^t i \, dt + v(0)$ and $v(0) = 0$,

$$\begin{aligned} v &= \frac{1}{2 \times 10^{-6}} \int_0^t 6e^{-3000t} \, dt \cdot 10^{-3} \\ &= \frac{3 \times 10^3}{-3000} e^{-3000t} \Big|_0^t = (1 - e^{-3000t}) \text{ V} \end{aligned}$$

Example 6.4

Determine the current through a $200\text{-}\mu\text{F}$ capacitor whose voltage is shown in Fig. 6.9.

Solution:

The voltage waveform can be described mathematically as

$$v(t) = \begin{cases} 50t \text{ V} & 0 < t < 1 \\ 100 - 50t \text{ V} & 1 < t < 3 \\ -200 + 50t \text{ V} & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$

Since $i = C dv/dt$ and $C = 200 \mu\text{F}$, we take the derivative of v to obtain

$$i(t) = 200 \times 10^{-6} \times \begin{cases} 50 & 0 < t < 1 \\ -50 & 1 < t < 3 \\ 50 & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$
$$= \begin{cases} 10 \text{ mA} & 0 < t < 1 \\ -10 \text{ mA} & 1 < t < 3 \\ 10 \text{ mA} & 3 < t < 4 \\ 0 & \text{otherwise} \end{cases}$$

Thus the current waveform is as shown in Fig. 6.10.

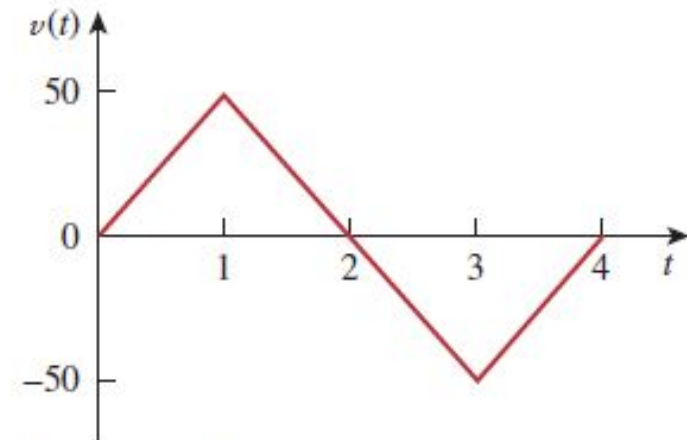


Figure 6.9

For Example 6.4.

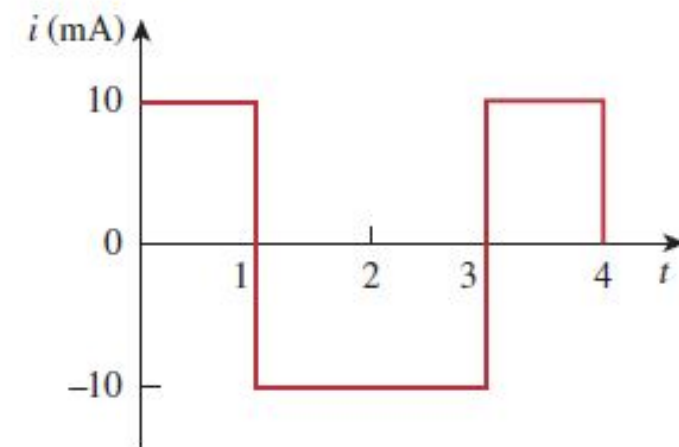


Figure 6.10

For Example 6.4.

An initially uncharged 1-mF capacitor has the current shown in Fig. 6.11 across it. Calculate the voltage across it at $t = 2$ ms and $t = 5$ ms.

Answer: 100 mV, 400 mV.

Practice Problem 6.4

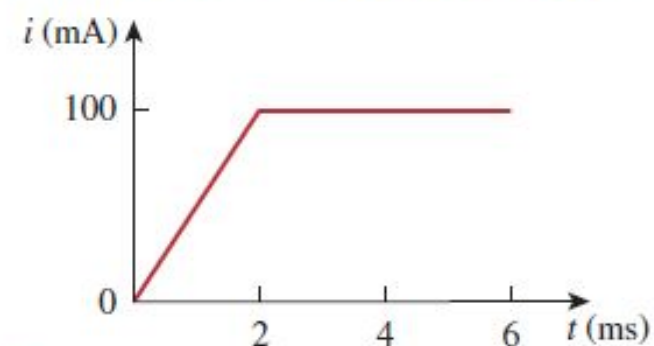


Figure 6.11
For Practice Prob. 6.4.

Example 6.5

Obtain the energy stored in each capacitor in Fig. 6.12(a) under dc conditions.

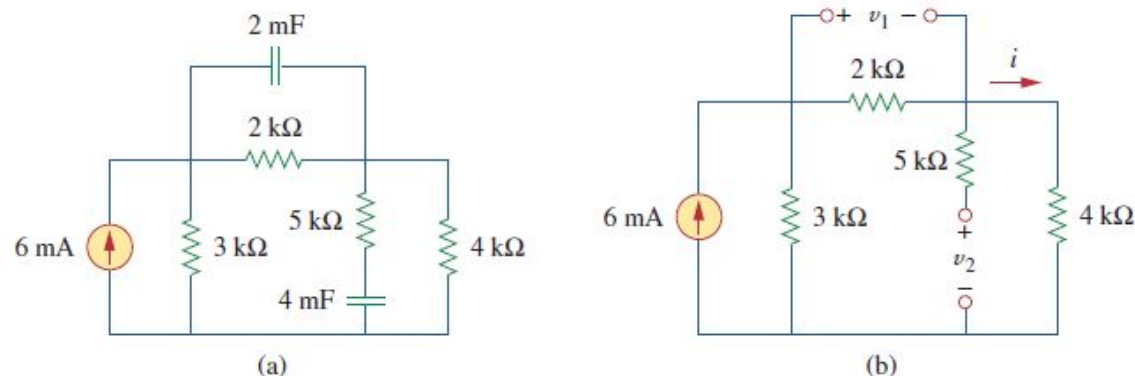


Figure 6.12
For Example 6.5.

Solution:

Under dc conditions, we replace each capacitor with an open circuit, as shown in Fig. 6.12(b). The current through the series combination of the 2-kΩ and 4-kΩ resistors is obtained by current division as

$$i = \frac{3}{3 + 2 + 4}(6 \text{ mA}) = 2 \text{ mA}$$

Hence, the voltages v_1 and v_2 across the capacitors are

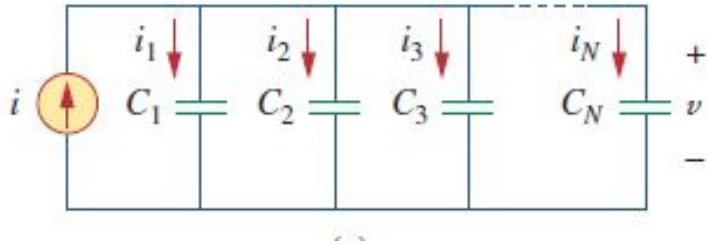
$$v_1 = 2000i = 4 \text{ V} \quad v_2 = 4000i = 8 \text{ V}$$

and the energies stored in them are

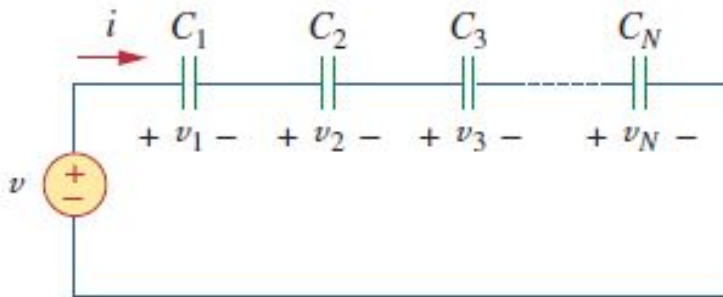
$$w_1 = \frac{1}{2}C_1v_1^2 = \frac{1}{2}(2 \times 10^{-3})(4)^2 = 16 \text{ mJ}$$

$$w_2 = \frac{1}{2}C_2v_2^2 = \frac{1}{2}(4 \times 10^{-3})(8)^2 = 128 \text{ mJ}$$

Series and Parallel Capacitors



$$C_{\text{eq}} = C_1 + C_2 + C_3 + \dots + C_N$$



$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N}$$

Example 6.7

For the circuit in Fig. 6.18, find the voltage across each capacitor.

Solution:

We first find the equivalent capacitance C_{eq} , shown in Fig. 6.19. The two parallel capacitors in Fig. 6.18 can be combined to get $40 + 20 = 60$ mF. This 60-mF capacitor is in series with the 20-mF and 30-mF capacitors. Thus,

$$C_{eq} = \frac{1}{\frac{1}{60} + \frac{1}{30} + \frac{1}{20}} \text{ mF} = 10 \text{ mF}$$

The total charge is

$$q = C_{eq}v = 10 \times 10^{-3} \times 30 = 0.3 \text{ C}$$

This is the charge on the 20-mF and 30-mF capacitors, because they are in series with the 30-V source. (A crude way to see this is to imagine that charge acts like current, since $i = dq/dt$.) Therefore,

$$v_1 = \frac{q}{C_1} = \frac{0.3}{20 \times 10^{-3}} = 15 \text{ V} \quad v_2 = \frac{q}{C_2} = \frac{0.3}{30 \times 10^{-3}} = 10 \text{ V}$$

Having determined v_1 and v_2 , we now use KVL to determine v_3 by

$$v_3 = 30 - v_1 - v_2 = 5 \text{ V}$$

Alternatively, since the 40-mF and 20-mF capacitors are in parallel, they have the same voltage v_3 and their combined capacitance is $40 + 20 = 60$ mF. This combined capacitance is in series with the 20-mF and 30-mF capacitors and consequently has the same charge on it. Hence,

$$v_3 = \frac{q}{60 \text{ mF}} = \frac{0.3}{60 \times 10^{-3}} = 5 \text{ V}$$

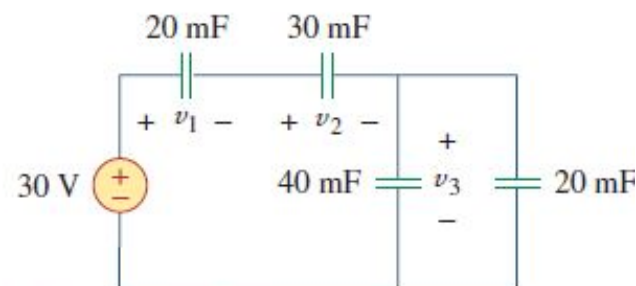


Figure 6.18

For Example 6.7.

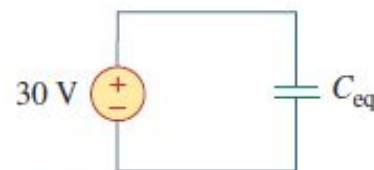


Figure 6.19

Equivalent circuit for Fig. 6.18.

Inductor

- An inductor is a passive element designed to store energy in its magnetic field.
- An inductor consists of a coil of conducting wire.

If current is allowed to pass through an inductor, it is found that the voltage across the inductor is directly proportional to the time rate of change of the current. Using the passive sign convention,

$$v = L \frac{di}{dt} \quad (6.18)$$

- Inductance (L) is the property whereby an inductor exhibits opposition to the change of current flowing through it, measured in henrys (H).

- The inductance of an inductor depends on its physical dimension and construction.

$$L = \frac{N^2 \mu A}{\ell}$$

- where N is the number of turns, ℓ is the length, A is the cross-sectional area, and μ is the permeability of the core.

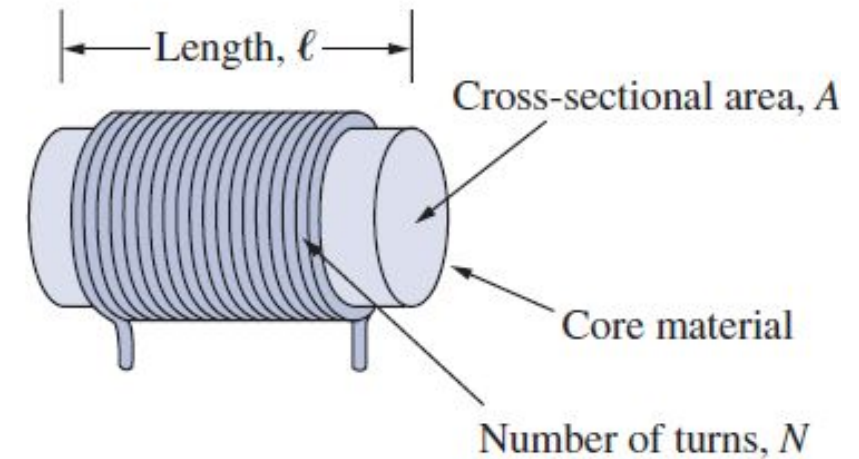


Figure 6.21

Typical form of an inductor.

The current-voltage relationship is obtained from Eq. (6.18) as

$$di = \frac{1}{L} v dt$$

Integrating gives

$$i = \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau \quad (6.20)$$

or

$$i = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0) \quad (6.21)$$

where $i(t_0)$ is the total current for $-\infty < t < t_0$ and $i(-\infty) = 0$. The idea of making $i(-\infty) = 0$ is practical and reasonable, because there must be a time in the past when there was no current in the inductor.

The inductor is designed to store energy in its magnetic field. The energy stored can be obtained from Eq. (6.18). The power delivered to the inductor is

$$p = vi = \left(L \frac{di}{dt} \right) i \quad (6.22)$$

The energy stored is

$$\begin{aligned} w &= \int_{-\infty}^t p(\tau) d\tau = L \int_{-\infty}^t \frac{di}{d\tau} i d\tau \\ &= L \int_{-\infty}^t i di = \frac{1}{2} Li^2(t) - \frac{1}{2} Li^2(-\infty) \end{aligned} \quad (6.23)$$

Since $i(-\infty) = 0$,

$$w = \frac{1}{2}Li^2 \quad (6.24)$$

We should note the following important properties of an inductor.

1. Note from Eq. (6.18) that the voltage across an inductor is zero when the current is constant. Thus,

An inductor acts like a short circuit to dc.

2. An important property of the inductor is its opposition to the change in current flowing through it.

The current through an inductor cannot change instantaneously.

Example 6.9

Find the current through a 5-H inductor if the voltage across it is

$$v(t) = \begin{cases} 30t^2, & t > 0 \\ 0, & t < 0 \end{cases}$$

Also, find the energy stored at $t = 5$ s. Assume $i(v) > 0$.

Solution:

Since $i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$ and $L = 5$ H,

$$i = \frac{1}{5} \int_0^t 30t^2 dt + 0 = 6 \times \frac{t^3}{3} = 2t^3 \text{ A}$$

The power $p = vi = 60t^5$, and the energy stored is then

$$w = \int p dt = \int_0^5 60t^5 dt = 60 \left. \frac{t^6}{6} \right|_0^5 = 156.25 \text{ kJ}$$

Alternatively, we can obtain the energy stored using Eq. (6.24), by writing

$$w|_0^5 = \frac{1}{2} Li^2(5) - \frac{1}{2} Li^2(0) = \frac{1}{2} (5)(2 \times 5^3)^2 - 0 = 156.25 \text{ kJ}$$

as obtained before.

Example 6.10

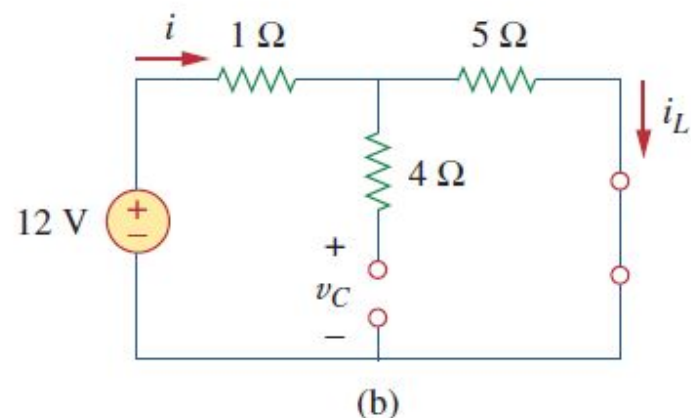
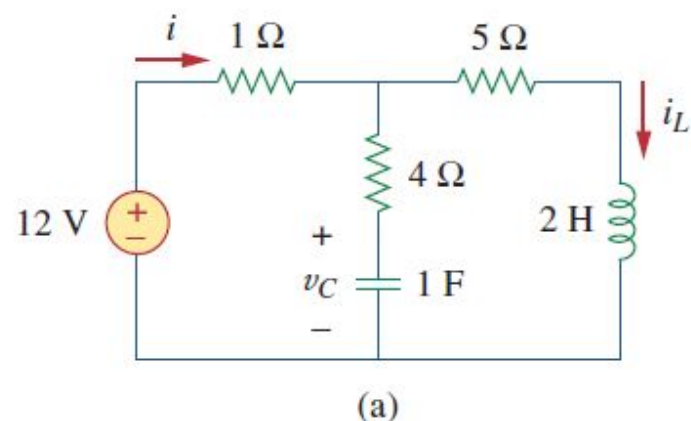


Figure 6.27

For Example 6.10.

Consider the circuit in Fig. 6.27(a). Under dc conditions, find: (a) i , v_C , and i_L , (b) the energy stored in the capacitor and inductor.

Solution:

(a) Under dc conditions, we replace the capacitor with an open circuit and the inductor with a short circuit, as in Fig. 6.27(b). It is evident from Fig. 6.27(b) that

$$i = i_L = \frac{12}{1 + 5} = 2 \text{ A}$$

The voltage v_C is the same as the voltage across the 5- Ω resistor. Hence,

$$v_C = 5i = 10 \text{ V}$$

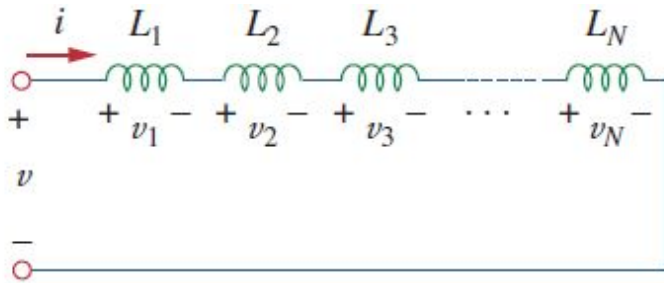
(b) The energy in the capacitor is

$$w_C = \frac{1}{2}Cv_C^2 = \frac{1}{2}(1)(10^2) = 50 \text{ J}$$

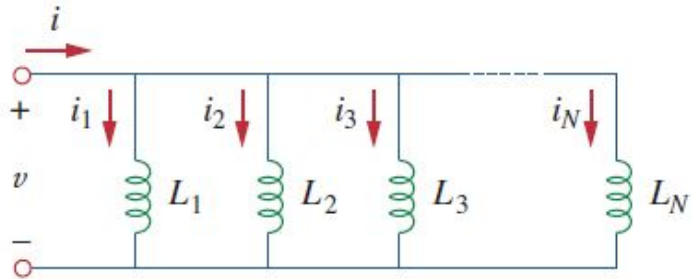
and that in the inductor is

$$w_L = \frac{1}{2}Li_L^2 = \frac{1}{2}(2)(2^2) = 4 \text{ J}$$

Series and Parallel Inductors



$$L_{eq} = L_1 + L_2 + L_3 + \dots + L_N$$



$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_N}$$

TABLE 6.1

Important characteristics of the basic elements.[†]

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v - i :	$v = iR$	$v = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$	$v = L \frac{di}{dt}$
i - v :	$i = v/R$	$i = C \frac{dv}{dt}$	$i = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$
p or w :	$p = i^2 R = \frac{v^2}{R}$	$w = \frac{1}{2} C v^2$	$w = \frac{1}{2} L i^2$
Series:	$R_{\text{eq}} = R_1 + R_2$	$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}$	$L_{\text{eq}} = L_1 + L_2$
Parallel:	$R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$	$C_{\text{eq}} = C_1 + C_2$	$L_{\text{eq}} = \frac{L_1 L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit

Example 6.12

For the circuit in Fig. 6.33, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find: (a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$; (c) $i_1(t)$ and $i_2(t)$.

Solution:

(a) From $i(t) = 4(2 - e^{-10t})$ mA, $i(0) = 4(2 - 1) = 4$ mA. Since $i = i_1 + i_2$,

$$i_1(0) = i(0) - i_2(0) = 4 - (-1) = 5 \text{ mA}$$

(b) The equivalent inductance is

$$L_{\text{eq}} = 2 + 4 \parallel 12 = 2 + 3 = 5 \text{ H}$$

Thus,

$$v(t) = L_{\text{eq}} \frac{di}{dt} = 5(4)(-1)(-10)e^{-10t} \text{ mV} = 200e^{-10t} \text{ mV}$$

and

$$v_1(t) = 2 \frac{di}{dt} = 2(-4)(-10)e^{-10t} \text{ mV} = 80e^{-10t} \text{ mV}$$

Since $v = v_1 + v_2$,

$$v_2(t) = v(t) - v_1(t) = 120e^{-10t} \text{ mV}$$

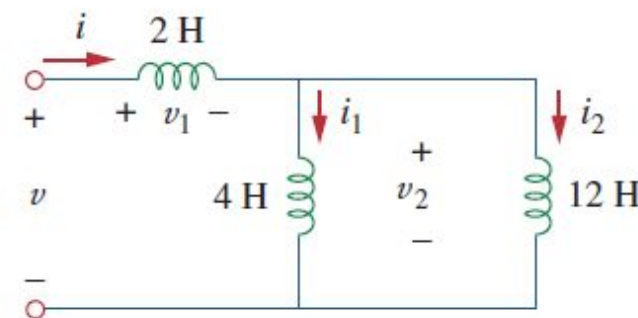


Figure 6.33

For Example 6.12.

(c) The current i_1 is obtained as

$$\begin{aligned} i_1(t) &= \frac{1}{4} \int_0^t v_2 dt + i_1(0) = \frac{120}{4} \int_0^t e^{-10t} dt + 5 \text{ mA} \\ &= -3e^{-10t} \Big|_0^t + 5 \text{ mA} = -3e^{-10t} + 3 + 5 = 8 - 3e^{-10t} \text{ mA} \end{aligned}$$

Similarly,

$$\begin{aligned} i_2(t) &= \frac{1}{12} \int_0^t v_2 dt + i_2(0) = \frac{120}{12} \int_0^t e^{-10t} dt - 1 \text{ mA} \\ &= -e^{-10t} \Big|_0^t - 1 \text{ mA} = -e^{-10t} + 1 - 1 = -e^{-10t} \text{ mA} \end{aligned}$$

Note that $i_1(t) + i_2(t) = i(t)$.

Practice Problem 6.12

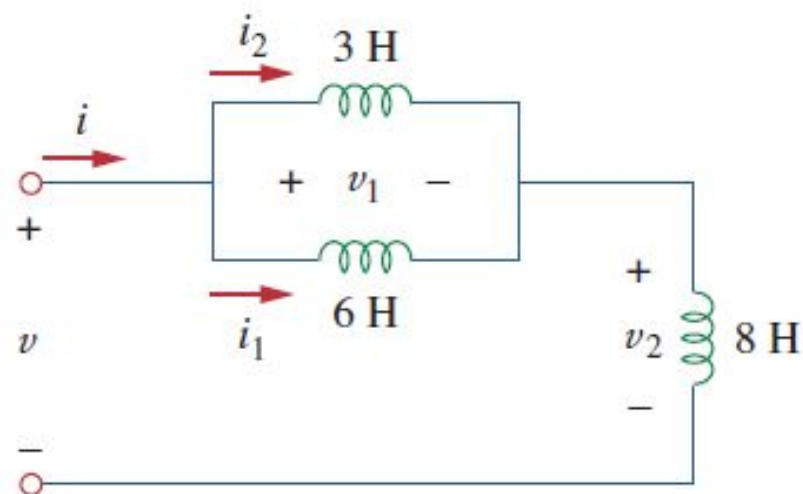


Figure 6.34

For Practice Prob. 6.12.

In the circuit of Fig. 6.34, $i_1(t) = 0.6e^{-2t}$ A. If $i(0) = 1.4$ A, find:
(a) $i_2(0)$; (b) $i_2(t)$ and $i(t)$; (c) $v_1(t)$, $v_2(t)$, and $v(t)$.

Answer: (a) 0.8 A, (b) $(-0.4 + 1.2e^{-2t})$ A, $(-0.4 + 1.8e^{-2t})$ A,
(c) $-36e^{-2t}$ V, $-7.2e^{-2t}$ V, $-28.8e^{-2t}$ V.